

IMMEDIATE OPERATIONAL USE

MODULAR ADDENDUM

Annex V

Sub-Surface Remediation & Joint Sovereignty Annex

"The Nuclear Dust Protocol"

Self-Contained · Incorporable by Reference

Version 1.0 — Emergency Stabilization Measure

Expires automatically upon completion of remediation or invocation of Article 4

All obligations in this Annex execute in active present tense.

No Party waits. Every trigger operates as a live operational mechanism.

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Preamble

Whereas the parties acknowledge that the relevant highly enriched uranium (HEU) is no longer in intact form but is physically entombed as dispersed particulate matter ("Nuclear Dust") beneath collapsed geological strata at previously declared military engineering sites;

Whereas no party accepts unilateral armed foreign excavation on its sovereign territory;

Whereas a verified, multilateral extraction mechanism is required to recover the Nuclear Dust to IAEA-certified standards while preserving Iranian surface sovereignty and excluding active-duty U.S. and Iranian military personnel from the remediation site;

The Parties adopt this Annex V as the binding operational mechanism governing extraction, verification, and financial sequencing for all Nuclear Dust remediation activities. This Annex treats every obligation as a live operational trigger, not a future promise.

Present-Tense Execution Principle: Every clause in this Annex executes in the active present tense. Consequences activate in real time. No Party waits for subsequent authorization once a condition is satisfied. Triggers fire automatically. Penalties accrue immediately. Releases occur upon verification, not upon political consensus.

Modularity Clause. This Annex V is self-contained and is incorporable into any existing or future agreement between the Parties through a single sentence of adoption. It does not amend or supersede conflicting provisions of any parent framework unless the Parties explicitly so agree. In the event of inconsistency, this Annex V governs with respect to Nuclear Dust remediation, dual-key sovereignty, financial micro-tranche sequencing, CANDOR governance, and emergency break-point protocols.

Article 1: Coordinated Access for Nuclear Dust Remediation (CANDOR)

1.1 CANDOR Composition and Governance

The Coordinated Access for Nuclear Dust Remediation consortium ("CANDOR") operates as the sole authorized international body for Nuclear Dust extraction. CANDOR comprises the following entities with defined roles:

Table V-1: CANDOR Composition and Roles

Entity	Role	Responsibilities
People's Republic of China	Co-Lead (Operational)	Site command coordination; extraction engineering oversight; dispute resolution between CANDOR members; liaison to Tehran and Washington
International Atomic Energy Agency (IAEA)	Co-Lead (Verification)	Radiological assay and mass balance certification; containerization standards enforcement; seal application and verification; real-time data feeds to all Parties
European Signatories (E3+: UK, France, Germany, EU representative)	Technical Operations	Specialized extraction equipment provision; environmental monitoring; waste containment engineering; technical training for non-military excavation crews
Russian Federation	Observer / Escrow Guarantor	Independent observation of CANDOR operations; escrow fund oversight for financial tranche releases; technical advisory on sub-surface recovery methodology

1.2 Personnel Exclusions and Access Rules

The following personnel restrictions apply at all CANDOR operational sites:

- a. No active-duty U.S. military personnel enter Iranian territory for CANDOR purposes. U.S. engineering expertise reaches the site only through third-party private contractors operating under IAEA badge authority and live video feed access for both Washington and Tehran.
- b. No active-duty Iranian military personnel enter the CANDOR operational perimeter during active extraction. Iranian Atomic Energy Organization (AEOI) technical observers maintain a presence in the site coordination center outside the active extraction zone.
- c. All CANDOR personnel carry IAEA-issued credentials. Iran's Ministry of Foreign Affairs receives the full personnel roster 72 hours before each rotation. Iran retains the right to deny entry to any individual on security grounds, with written justification provided within 24 hours.
- d. Maximum CANDOR non-Iranian personnel at any given time: 85 individuals. This cap includes Chinese operational staff, European technical personnel, IAEA inspectors, and Russian observers. The

cap excludes AEOI technical observers and Iranian site security personnel.

1.3 Extraction Mandate and Standards

CANDOR is required to physically excavate, recover, and containerize a minimum of 99.7% of the estimated 440.9 kg of 60%-enriched HEU existing as Nuclear Dust. The IAEA certifies recovery percentage through mass spectrometry and radiological assay at each extraction milestone.

The recovery target breaks down as follows:

Table V-2: Recovery Milestones and Verification Standards

Milestone	Recovery Target	IAEA Verification Method	Timeline
Initial Site Survey	Confirm dust presence and distribution	Core drilling assay; gamma spectroscopy	Days 1-14
Phase I Extraction	25% recovery (110.2 kg)	Real-time mass balance; container seal verification	Days 15-60
Phase II Extraction	75% recovery (330.7 kg cumulative)	Continuous assay; environmental contamination monitoring	Days 61-120
Final Containerization	99.7% recovery (439.6 kg cumulative)	Final mass spectrometry; container certification; chain-of-custody documentation	Days 121-180

1.4 Operational Security Protocols

CANDOR operates under the following security framework:

- A 5-kilometer exclusion zone surrounds each operational site. Iranian police maintain the outer perimeter. CANDOR security maintains the inner perimeter.
- All extraction equipment enters through a single designated logistics corridor, inspected by joint Iranian-IAEA teams.
- Every container of recovered Nuclear Dust receives an IAEA tamper-evident seal with unique serial number at the point of extraction.
- Live encrypted video feeds transmit to monitoring stations in Tehran, Washington, Beijing, Vienna (IAEA), and Moscow simultaneously. No feed interruption exceeds 15 minutes without automatic escalation to the Joint Secretariat.

Article 2: Dual-Key Sovereignty Provisions

2.1 Digital Key Architecture

Each excavation shift requires two digital authorization keys operating in tandem:

- a. **Iran Key:** The AEOI Director General (or designated deputy) holds one digital key. This key authorizes the shift from the Iranian sovereignty perspective. Without this key, no extraction equipment activates.
- b. **IAEA Key:** The IAEA Director General (or designated senior inspector on-site) holds the second digital key. This key certifies that radiological safety conditions, extraction protocols, and verification standards are met. Without this key, no extraction equipment activates.

Both keys are required simultaneously for every shift. Either key holder withdraws authorization at any time, halting extraction immediately. A key withdrawal triggers automatic notification to all Parties within 15 minutes via the Shared Digital Registry.

2.2 U.S. Personnel Access Restrictions

The following access rules apply specifically to U.S. nationals:

- a. No U.S. national enters Iranian territory for CANDOR purposes without explicit, time-limited AEOI visa approval. Each visa is valid for a single shift rotation (maximum 12 hours) and a specific defined zone.
- b. The total number of U.S. nationals present at any given time does not exceed 30 individuals. All 30 are non-uniformed civilian contractors operating under IAEA badge authority.
- c. U.S. nationals perform no operational role in direct extraction. Their permitted activities include: remote engineering consultation via secure video link; technical data review in the CANDOR coordination center; and observation of IAEA verification procedures.
- d. Iran receives the full name, passport number, and professional biography of each U.S. national 7 days before visa issuance. Iran is required to approve or deny each application within 72 hours. A denial requires written justification; a failure to respond within 72 hours constitutes automatic approval.

2.3 Iranian Surface Sovereignty

Iran retains full surface sovereignty over all CANDOR operational sites. This sovereignty includes:

- Ultimate authority over all personnel entering Iranian territory for CANDOR purposes
- Control of the outer security perimeter and all access roads to the operational sites
- The right to suspend CANDOR operations for up to 48 hours in response to verified security threats, with automatic notification to all Parties and the Joint Secretariat
- Jurisdiction over any criminal or civil matters arising outside the CANDOR operational perimeter

2.4 Easement Terms and Revocation

CANDOR operates under a limited, revocable easement for access to the three defined collapsed zones. The GPS-locked coordinates of these zones are attached as Annex V-2A.

The easement terms are:

- **Scope:** The easement permits only sub-surface extraction, containerization, and environmental remediation activities. It does not authorize permanent construction, military presence, or infrastructure development unrelated to Nuclear Dust recovery.
- **Duration:** The easement remains valid for the duration of CANDOR active operations, up to a maximum of 180 days from the date of Annex V entry into force.
- **Revocation:** Iran revokes the easement with 72 hours' written notice to the Joint Secretariat and all CANDOR members. Revocation is not permitted during an active extraction shift but takes effect at the conclusion of the current shift. Upon revocation, Article 4 (Emergency Break Points) governs the financial and operational transition.
- **Renewal:** The Parties extend the easement by mutual written agreement. No single Party extends the easement unilaterally.

Article 3: Financial Micro-Tranche Lockstep ("Dust-for-Dollars")

3.1 General Architecture

The disputed \$12 billion in frozen Iranian assets held in escrow is not released upfront. Instead, asset release occurs in strict micro-tranches tied to verified extraction milestones. The Swiss facilitator administers the escrow account and executes releases within 24 hours of IAEA certification.

3.2 Initial Assay Tranche

Upon IAEA verification of Nuclear Dust presence through on-site radiological assay, the Swiss facilitator releases \$500 million from the frozen escrow. This release occurs within 24 hours of IAEA certification. The assay verifies that a minimum of 400 kg of HEU-equivalent dispersed material is present at the designated sites.

3.3 Per-Ton Extraction Tranches

For every metric ton of Nuclear Dust extracted (measured in 10-ton cumulative tranches), the Swiss facilitator releases \$200 million from the escrow. The release triggers are:

Table V-3: Per-Ton Extraction Tranche Schedule

Cumulative Extraction	Tranche Release	Cumulative Released	Verification Trigger
10 tons	\$200M	\$700M (\$500M + \$200M)	IAEA mass balance confirms 10 tons containerized
20 tons	\$200M	\$900M	IAEA mass balance confirms 20 tons containerized
30 tons	\$200M	\$1.1B	IAEA mass balance confirms 30 tons containerized
40 tons	\$200M	\$1.3B	IAEA mass balance confirms 40 tons containerized
50 tons	\$200M	\$1.5B	IAEA mass balance confirms 50 tons containerized
60 tons	\$200M	\$1.7B	IAEA mass balance confirms 60 tons containerized
70 tons	\$200M	\$1.9B	IAEA mass balance confirms 70 tons containerized
80 tons	\$200M	\$2.1B	IAEA mass balance confirms 80 tons containerized

90 tons	\$200M	\$2.3B	IAEA mass balance confirms 90 tons containerized
100 tons	\$200M	\$2.5B	IAEA mass balance confirms 100 tons containerized

Note: The 100-ton figure represents total excavated material including soil matrix and debris. The 440.9 kg HEU equivalent is recovered from within this matrix. The IAEA certifies both total tonnage and HEU concentration per tranche.

3.4 Final Containerization Payment

Upon IAEA certification that 99.7% of the Nuclear Dust is containerized and ready for escorted transport to neutral-country storage, the Swiss facilitator releases the remaining balance of the \$12 billion escrow, minus the amounts already released through prior tranches.

In addition to the remaining balance, a \$1 billion completion bonus is paid and split 50/50 between:

- **Iran's Central Bank** (50% = \$500 million), deposited as sovereign liquid funds with no restriction; and
- **A designated humanitarian escrow** (50% = \$500 million), administered by UN OCHA for food and medicine procurement benefiting the civilian population of the Gaza Strip and southern Lebanon, in coordination with the Regional Stability Reconstruction Credits under Article 10 of the parent framework.

3.5 Escrow and Verification Mechanism

The Swiss facilitator maintains the escrow account with the following operational rules:

- The escrow account is domiciled at the Central Bank of Oman under sovereign immunity protections.
- IAEA certification is the sole release trigger. No political entity overrides an IAEA-certified release.
- Every release generates an automatic entry in the Shared Digital Registry (Annex U, Article 5), timestamped and visible to all Parties.
- If IAEA certification is delayed beyond 72 hours due to technical review, the Swiss facilitator holds the funds in a pending-release sub-account earning interest at the Omani central bank rate.
- Russia, as escrow guarantor, receives real-time notification of every deposit, release, and balance change. Russia does not hold a release key but serves as an independent balance auditor.

Article 4: Emergency Break Points

4.1 90% Recovery Threshold

If CANDOR determines after 45 days of continuous operation that extraction below 90% recovery is physically impossible due to geological conditions, radiation levels, or structural collapse beyond the engineering capacity of available equipment, the Parties revert automatically to in-situ monitored entombment.

The determination requires:

- Unanimous certification by the CANDOR technical leadership (China co-lead, IAEA co-lead, and European technical lead)
- Independent verification by a second-party engineering assessment team (comprising non-CANDOR specialists from neutral countries, selected by the Joint Secretariat)
- Written confirmation that no alternative extraction methodology exists that is capable of achieving 90% recovery within the remaining operational window

4.2 In-Situ Entombment Protocol

Upon invocation of the 90% threshold break point, CANDOR transitions to the in-situ monitored entombment protocol:

- a. All remaining Nuclear Dust is left in place and sealed using permanent IAEA-approved containment barriers.
- b. The IAEA installs perpetual camera surveillance and continuous radiation monitoring at all entombment sites. Data feeds transmit in real time to Tehran, Washington, Beijing, Vienna, and Moscow.
- c. Tamper-evident seals are applied at every access point. Any breach of these seals triggers automatic notification to all Parties within 5 minutes and constitutes a material breach of this Annex V.
- d. The IAEA conducts quarterly physical inspections of the entombment sites. Annual comprehensive reviews assess seal integrity, radiation levels, and structural stability.
- e. No further excavation occurs at the entombment sites without unanimous consent of all Parties and IAEA safety certification.

4.3 50% Release / 50% Hold Split

Upon invocation of the entombment protocol, the Swiss facilitator automatically executes the following split of the frozen \$12 billion:

- **50% Released (\$6 billion):** Released to Iran's Central Bank within 48 hours of IAEA entombment certification. This amount compensates Iran for the verified extraction work completed and acknowledges the permanent IAEA monitoring commitment.

- **50% Held (\$6 billion):** Remains in escrow until a joint U.S.-Iranian technical commission confirms entombment integrity. The commission mediates through China and comprises three U.S. technical experts, three Iranian technical experts, and one Chinese chair.

4.4 Joint Technical Commission Review

The joint U.S.-Iranian technical commission (mediated by China) convenes within 30 days of entombment protocol invocation. The commission is required to:

- a. Review all IAEA entombment data, seal integrity reports, and radiation monitoring logs
- b. Conduct an independent on-site inspection within 60 days of commission formation
- c. Issue a binding determination on entombment integrity within 90 days of commission formation
- d. If the commission confirms entombment integrity by majority vote (5 of 7 members), the Swiss facilitator releases the remaining \$6 billion to Iran's Central Bank within 24 hours
- e. If the commission determines that entombment integrity is compromised, the Parties negotiate remedial measures. During remediation, the held \$6 billion remains in escrow. No interest accrues to either Party during the remediation period

Article 5: Verification and Telemetry Architecture

5.1 Real-Time Monitoring Infrastructure

CANDOR maintains the following real-time monitoring infrastructure at every operational site:

Table V-4: Telemetry Infrastructure and Data Feeds

System	Function	Feed Recipients	Latency
Gamma Spectroscopy Array	Continuous radiological monitoring of extraction zone	IAEA Vienna; AEOI Tehran; U.S. DOE	Real-time (5-second refresh)
Encrypted Video Network	Visual monitoring of all extraction and containerization activities	All Parties; Joint Secretariat	Live (15-second max interruption)
Container Seal GPS Tracker	Location and integrity tracking of every sealed container	IAEA; Swiss facilitator; Russian escrow auditor	1-minute position updates
Environmental Sensor Grid	Air, soil, and groundwater contamination monitoring	Iran EPA; IAEA; European technical team	15-minute sampling cycles
Mass Balance Ledger	Real-time tracking of extracted material weight and HEU concentration	All Parties; Shared Digital Registry	Updated per shift (every 12 hours)

5.2 IAEA Mass Balance Verification

The IAEA certifies the mass balance for each extraction tranche through the following procedure:

- Pre-extraction baseline measurement of the extraction zone volume and estimated HEU dispersion
- Per-shift weighing of all extracted material with certified industrial scales (calibrated daily)
- Per-shift gamma spectroscopy assay to determine HEU concentration in the extracted matrix
- Post-shift reconciliation: total weight x HEU concentration = HEU mass recovered
- Independent verification by a second IAEA inspector team using separate equipment
- Publication of the certified mass balance to the Shared Digital Registry within 6 hours of shift completion

5.3 Data Sharing Symmetry

All telemetry data is shared symmetrically. No Party receives preferential access, earlier notification, or exclusive data feeds. Every data stream transmits simultaneously to all designated recipients. Any data asymmetry constitutes a Tier 2 Material Disruption under Annex T of the parent framework.

Article 6: Implementation Timeline and Phases

6.1 Phase I: Site Preparation and Verification (Days 1-14)

During Phase I, the following activities occur:

- Day 1: CANDOR advance team arrives in Iran; equipment staging begins at designated logistics hub
- Day 3: IAEA core drilling team initiates subsurface assay at all three collapsed zones
- Day 7: Preliminary assay results confirm Nuclear Dust presence and distribution; AEOI issues initial shift authorizations
- Day 10: Dual-key digital infrastructure is installed and tested; first joint key-holding drill occurs
- Day 14: Phase I report issued to all Parties; IAEA certifies dust presence for initial \$500M tranche release; extraction equipment positioned at Zone 1

6.2 Phase II: Active Extraction (Days 15-120)

During Phase II, CANDOR conducts continuous extraction operations:

- Days 15-30: Zone 1 primary extraction; target 15% total recovery (66.1 kg HEU equivalent)
- Days 31-45: Zone 1 completion and Zone 2 initiation; 90% recovery threshold assessment occurs
- Days 46-75: Zone 2 primary extraction; target 50% cumulative recovery (220.5 kg)
- Days 76-105: Zone 3 initiation; target 80% cumulative recovery (352.7 kg)
- Days 106-120: Final extraction push across all zones; target 95% cumulative recovery (418.9 kg)

6.3 Phase III: Containerization and Transfer (Days 121-180)

During Phase III, all remaining material is containerized and prepared for transfer:

- Days 121-140: Final 4.7% recovery extraction; all material consolidated into IAEA-certified containers
- Days 141-150: Final IAEA mass spectrometry and certification of 99.7% recovery
- Days 151-160: Container convoy preparation; security escort coordination with Iranian police and CANDOR security
- Days 161-175: Escorted transport of containers to neutral-country storage facility (Oman or Türkiye, by mutual agreement)
- Days 176-180: Final storage confirmation; IAEA chain-of-custody documentation completed; remaining escrow balance and \$1B completion bonus released

6.4 Extension Protocol

If CANDOR determines before Day 150 that the 180-day timeline is insufficient to achieve 99.7% recovery, CANDOR submits a written extension request to all Parties. The extension requires mutual written

agreement and is limited to a single 60-day extension. No extension beyond 240 total days is permitted. If 99.7% recovery is not achieved by Day 240, Article 4 (Emergency Break Points) governs automatically.

Article 7: Relationship to Existing Frameworks

7.1 Conjunction with Annex U

This Annex V operates in conjunction with Annex U (Technical Demarcation and Stabilization Addendum). Any territorial breach affecting CANDOR operational sites or logistics corridors triggers the Automatic Breach mechanism under Annex U, Article 3. The Shared Digital Registry under Annex U, Article 5 records all Annex V telemetry data, financial tranche releases, and mass balance certifications.

7.2 Conjunction with Annex M

This Annex V recognizes the territorial baselines established in Annex M (Territorial Baseline and Ceasefire Demarcation Protocol). CANDOR operational sites are located within the Current Operational Footprint as defined in Annex M, Article 1.1. No CANDOR activity alters the territorial baselines established in Annex M.

7.3 Integration with the Parent Framework

This Annex V integrates with the parent framework (The Final Steps Accord, Phase I) through the following linkages:

- Article 1 (HEU Custody Exchange Protocol): Annex V provides the physical extraction mechanism for the HEU addressed in the parent framework's custody architecture
- Article 8 (Neutral Escrow): The \$12 billion escrow is administered through the same Swiss facilitator architecture established in the parent framework
- Article 10 (Regional Stability Reconstruction Credits): The \$500 million humanitarian portion of the completion bonus flows through the UN OCHA mechanism established under Article 10
- Article 12 (Distributed Guarantor Matrix): China, Russia, the EU, and Oman serve as guarantors for Annex V operations, consistent with their guarantor roles in the parent framework

7.4 No Amendment of Parent Framework

This Annex V does not amend the parent framework. It is a modular, self-contained addendum that addresses a specific operational gap (Nuclear Dust remediation) not fully covered by existing annexes. All provisions of the parent framework remain operative and are supplemented, not replaced, by this Annex V.

Article 8: Incorporation by Reference

This Annex V is incorporable into any existing or future agreement through the following sentence:

"The Parties adopt Annex V (Sub-Surface Remediation & Joint Sovereignty Annex) as an operative provision of this Agreement. In the event of any inconsistency, Annex V governs with respect to Nuclear Dust remediation, CANDOR governance, dual-key sovereignty, financial micro-tranche sequencing, verification telemetry, and emergency break-point protocols."

No amendment of the parent framework is required. This Annex V enters into force upon signature by both Parties or upon provisional application by exchange of diplomatic notes.

Article 9: Provisional Application

9.1 Provisional Application Period

The Parties apply this Annex V provisionally for 90 days, pending formal incorporation. Provisional application is effected by exchange of diplomatic notes between the United States and the Islamic Republic of Iran, with notification to the Joint Secretariat.

9.2 Operative Binding During Provisional Application

During provisional application, all provisions of this Annex V are operative and binding, including: the full CANDOR extraction mandate; the dual-key sovereignty mechanism; the Dust-for-Dollars micro-tranche schedule; and the emergency break-point protocols.

9.3 Termination of Provisional Application

Either Party terminates provisional application with 14 days' written notice to the Joint Secretariat and all guarantor states. Termination of provisional application does not affect:

- Extraction work completed during provisional application
- Financial tranches already released and paid
- IAEA certifications already issued
- Breaches that occurred during provisional application

Note: The 14-day notice period allows for orderly transition without leaving active operations ungoverned. CANDOR operations wind down in an orderly manner during the notice period, with priority given to securing all extracted material and maintaining seal integrity at all sites.

Signature Block

For the Government of the Islamic Republic of Iran (as host sovereign and affected Party):

Name:

Title:

Date:

For the United States of America (as endorsing Party):

Name:

Title:

Date:

For the People's Republic of China (as CANDOR Co-Lead):

Name:

Title:

Date:

For the International Atomic Energy Agency (as Verification Co-Lead):

Name:

Title:

Date:

Witnessed by the Joint Secretariat (Muscat), the Russian Federation (Escrow Guarantor), and the European Union (E3+):

Signature:

Date:

Annex V — Sub-Surface Remediation & Joint Sovereignty Annex — Version 1.0

All obligations execute in active present tense — Expires upon completion of remediation or invocation of Article 4 entombment protocol